

Computer Science & IT

Database Management System



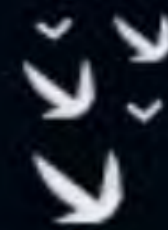
**File organization
And
Indexing**

Lecture No. 02



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Recap of Previous Lecture



✓
Topic

Database, File and Records

Topic

Organization of Records



Topics to be Covered



- ✓ **Topic** Index file
- ✓ **Topic** IO cost with indexing
- ✓ **Topic** Categories of index
- Topic** Types of index





Topic : IO cost without indexing

★ Let "N" is the total number of disk blocks required to store the given database file,

(i) Worst case IO cost = N {search will be linear search}
(When file is un-ordered)

(ii) Worst Case IO Cost = $\lceil \log_2 N \rceil$ {If file is ordered then}
(When file is ordered) {Binary search is possible}

★ Note:- If database is too large, then "N" will also be a large value and in that case $\lceil \log_2 N \rceil$ will also be a significant value
∴ We must do something to reduce the IO Cost (Hence concept of indexing will be used)

Ordered file

vs

Unorder file { Also known as Heap file }

① Binary search is possible

② Insertion of a new record in an ordered file will be a costly operation. (because record must be stored at appropriate position in file)

① Binary search is not possible

② Insertion of a new record will be easy. { We can simply insert the record in the last block of the file }



Topic : Index file

Index files are used to reduce the IO cost.



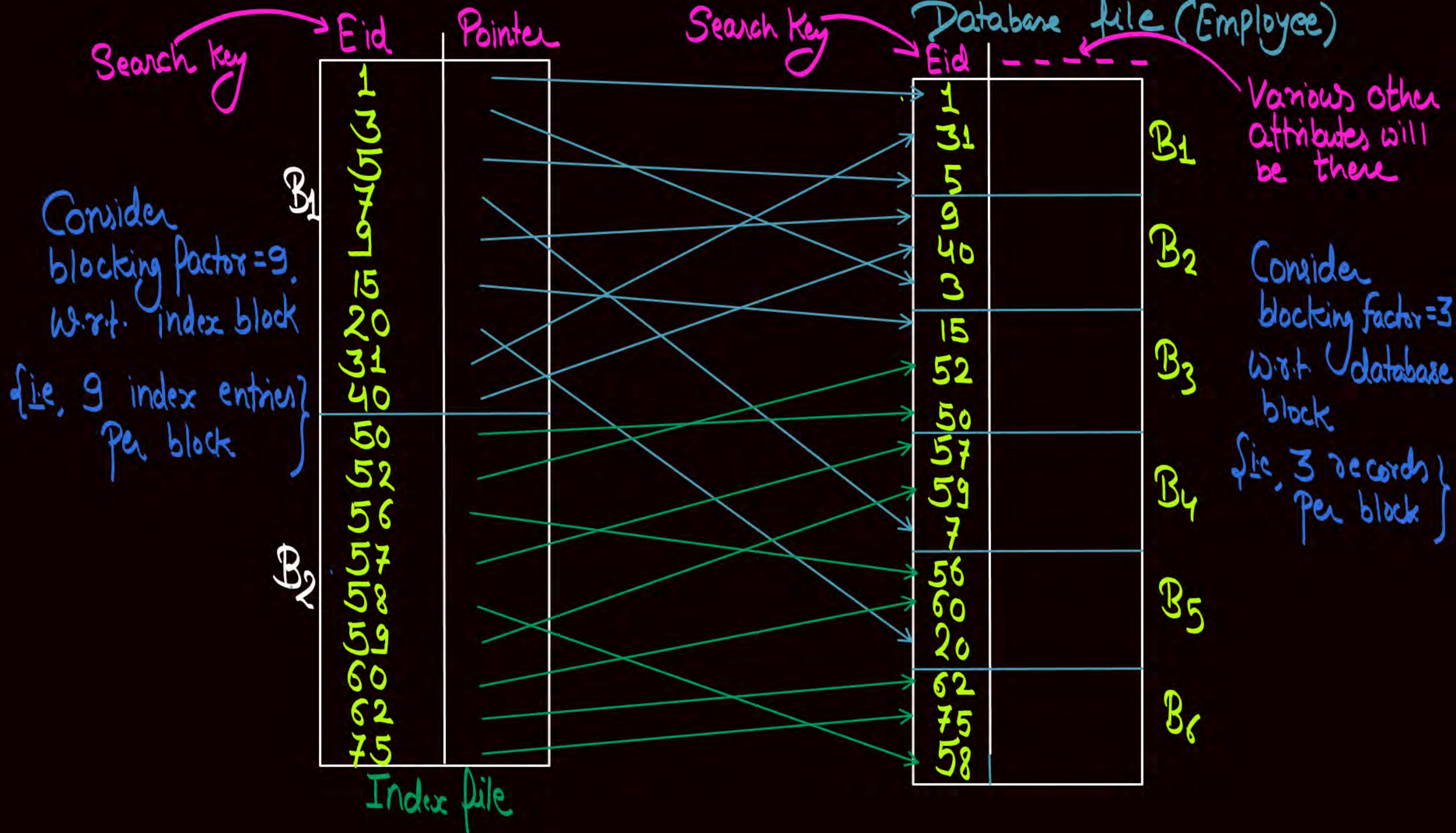
Each entry in the index file contains only two fields

- ① Search key attribute value
- ② Pointer pointing to the record / disk block corresponding to the search key value.

Index file Entry = $\langle \text{Search Key value, Pointer information} \rangle$

in general, $\underbrace{\text{Size of index file entry}}_{\text{Contains only two fields}} < \underbrace{\text{Size of each record in DB file}}_{\text{Contains multiple fields in each record}}$

→ Index file is also stored in the blocks of the same disk.
∴ $\text{Size of disk blocks used to store index file} = \text{Size of disk blocks used to store the records of the database file}$



- ① Index file entry = $\langle \text{Search Key value}, \text{Pointer information} \rangle$
- ② Index file entry size = $(\text{Search key attribute size} + \text{Pointer size})$
- ③ Database Record Size = $(\text{Summation of the sizes required to store each attribute of the database file})$
- ④ Disk block size is same for database file and index file
- ⑤ Blocking factor (w.r.t index block) = $\frac{\text{Disk block size}}{\text{Index file entry size}}$
- ⑥ Blocking Factor (w.r.t. database block) = $\frac{\text{Disk block size}}{\text{Record size}}$
- ⑦ In general, Index file Entry size < database file record size
or By ④, ⑤, ⑥, ⑦
- ⑧ $\rightarrow$ Blocking factor (w.r.t index block) > Blocking Factor (w.r.t. database block)

$$\begin{aligned} \textcircled{9} \text{ No. of disk blocks required to store index file} &= \frac{\text{Total no. of entries in the index file}}{\text{Number of index entries per block}} \\ &= \frac{\text{Total no. of entries in the index file}}{\text{Blocking factor of index block}} \end{aligned}$$

$$\begin{aligned} \textcircled{10} \text{ No. of disk blocks required to store database file} &= \frac{\text{Total no. of records in the database file}}{\text{Number of records per block}} \\ &= \frac{\text{Total no. of records in the database file}}{\text{Blocking factor of database block}} \end{aligned}$$

⑪ In general,

No. of entries in
index file

$<$

wrt. sparse index

$\leq$

No. of records in
database file

$=$

wrt. dense index.

⑫ By ⑧, ⑨, ⑩ & ⑪

Number of disk blocks
required to store index file

$<<$

Number of disk blocks
required to store database file

Note:-

In general, Index file is ordered based on
Search Key

→ ∴ We can always perform
binary search in index file



Topic : IO cost with indexing

Let 'M' is the number of disk blocks required to store index file, and N is the number of disk blocks required to store the database file, then in general, $M \ll N$

$$\text{Worst case IO Cost with index file} = \lceil \log_2 M \rceil + 1$$

(to search for an entry in the index file w.r.t. given search key)

To access the block of database file, that contains the record corresponding to the search key value



Topic : Categories of Index

There are two categories of the indexes

- ① Dense index :- If we maintain an entry in the index file for each record of the database file, then it is called dense index.

i.e;
$$\begin{array}{l} \text{Number of entries in index file} \\ \text{w.r.t. dense index} \end{array} = \text{No. of records in the database file}$$



Topic : Categories of Index

There are two categories of the indexes

② Sparse index :- For a collection of records in the database file we maintain only one entry in index file, then it is sparse index

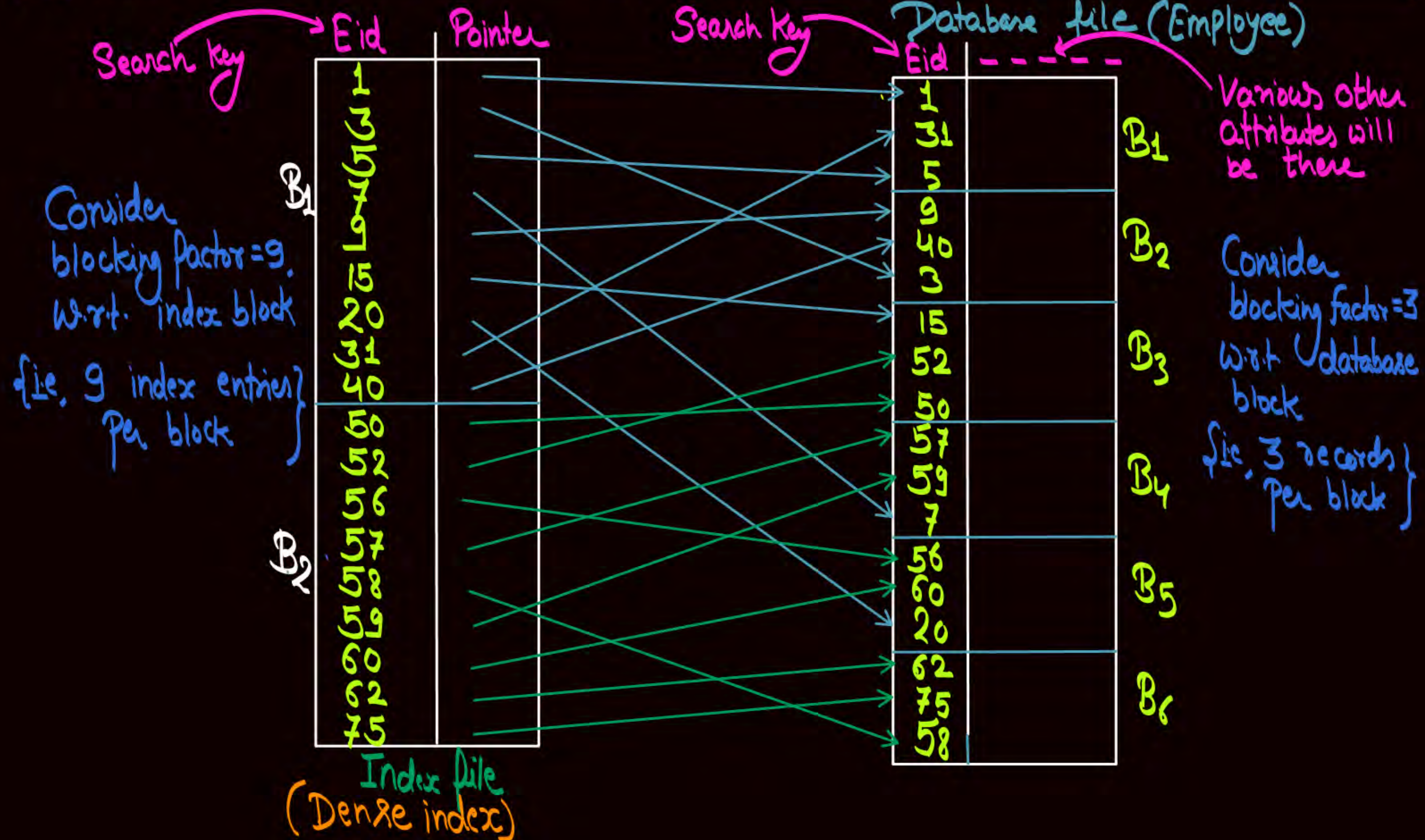
i.e; $\text{Number of entries in index file} < \text{No. of records in the database file}$
w.r.t. Sparse index

* If index is sparse index, and nothing else is specified in the question about the no. of entries in sparse index, then in general 1 index file entry per block of database file

i.e, $\text{No. of entries in Sparse index file} = \text{No. of disk blocks required to store database file}$

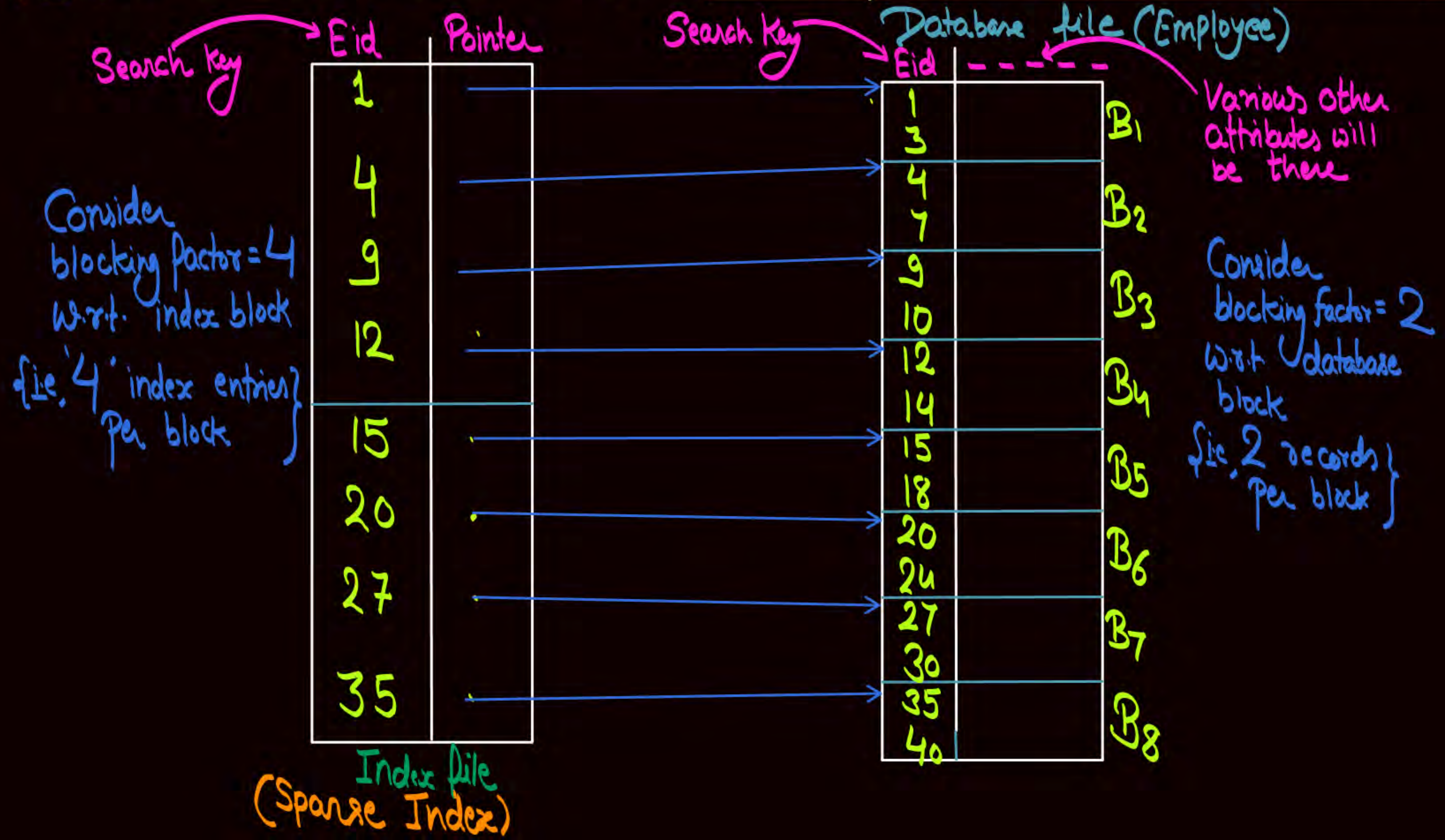
* Example of Dense index file: -

In general, dense index is created for unordered file



→ We can create dense index for ordered file as well, but generally it is created for unordered file

* Example of Sparse index file: - Sparse index can be created only for ordered files (Not possible for unordered file)



Note:-

① If file is ordered based on search key attribute values, then both sparse as well as dense index are possible

② If file is unordered based on search key attribute values, then only dense index is possible on that search key

{ sparse index is never possible for an unordered file }

Q:- Consider the following specifications.

Disk block size = 1000 Bytes

Record size = 100 Bytes

Search key attribute size = 12 Bytes

Record/Block pointer size = 8 Bytes

Let total no. of records in the database file = 10,000

- (1) How many disk blocks are required to store the dense index file on the above database file
- (2) How many disk blocks are required to store the sparse index file on the above database file

(1) No. of disk blocks required to store dense index file

$$= \frac{\text{No. of entries needs to be stored in dense index}}{\text{No of entries per disk block}}$$

$$= \frac{\text{No. of records in database file}}{\text{Blocking factor w.r.t index block}}$$

$$= \frac{10,000}{\left(\frac{\text{Block size}}{\text{index file entry size}} \right)} = \frac{10,000}{\left(\frac{1000}{(12+8)} \right)}$$

$$= \frac{10,000 \times 20}{1000}$$

$$\boxed{= 200}$$

② No. of disk blocks required to store Sparse index file =

$$= \frac{\text{No. of entries needs to be stored in Sparse index}}{\text{No. of entries per disk block}} \cdot \frac{\text{No. of disk blocks required to store Database file}}{\text{Blocking factor w.r.t index block}}$$

No. of disk blocks required to store database file =

$$= \frac{\text{No. of records needs to be stored}}{\text{No. of records per block}}$$

$$= \frac{10,000}{\text{Blocking factor w.r.t database block}}$$

$$= \frac{10,000}{\left(\frac{\text{Block size}}{\text{Record size}}\right)} = \frac{10,000}{\left(\frac{1000}{100}\right)} = 1000$$

$$= \frac{1000}{\left(\frac{\text{Block size}}{\text{index file entry size}}\right)} = \frac{1000}{\left(\frac{1000}{(12+8)}\right)}$$

$$= \boxed{20} \quad \underline{\underline{\text{Ans}}}$$



2 mins Summary



✓ **Topic**

Index file

✓ **Topic**

IO cost with indexing

✓ **Topic**

Categories of index

Topic

Types of index

THANK - YOU